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RESEARCH ARTICLE

Unsteady fluid–structure interactions of membrane airfoils at low Reynolds numbers

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Abstract Membrane wings are used both in nature and small aircraft as lifting surfaces. Separated flows are common at low Reynolds numbers and are the main sources of unsteadiness. Yet, the unsteady aspects of the fluid–structure interactions of membrane airfoils are largely unknown. An experimental study of unsteady aerodynamics of two-dimensional membrane airfoils at low Reynolds numbers has been conducted. Measurements of membrane shape with a high-speed camera were complemented with the simultaneous measurements of unsteady velocity field with a high frame-rate particle image velocimetry system and flow visualization. Vibrations of the membrane and mode shapes were investigated as a function of angle of attack and free stream velocity. While the mean membrane shape is not very sensitive to angle of attack, the amplitude and mode of the vibrations of the membrane depend on the relative location and the magnitude of the unsteadiness of the separated shear layer. The results indicate strong coupling of unsteady flow with the membrane oscillations. There is evidence of coupling of membrane oscillations with the vortex shedding in the wake, in particular, for the post-stall incidences. Comparison of rigid (but cambered) and flexible membrane airfoils shows that the flexibility might delay the stall. Hence this is a potential passive flow control method using flexibility in nature and engineering applications.

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1 Introduction

Flying mammals such as bats have thin compliant wings as lifting surfaces. These animals with flexible membranes fly at low Reynolds numbers at moderate to high angles of attack, and exhibit high maneuverability. It was suggested that bat flight might be more efficient than that of large insects or small birds of comparable size (Winter and von Helversen 1998). Flying and gliding mammals are well known for flight capabilities and agility (Bishop 2006). The skin of the bat wing is known to exhibit substantial changes in shape and camber throughout the wingbeat cycle (Swartz et al. 2007) while the effects of multiple joints and anisotropic membrane stiffness across the wing (Swartz et al. 1996) are not understood.

Membrane wings are used in many engineering applications including parachutes, microlight, paraglider and hang glider wings, yacht sails, and wings of small unmanned air vehicles known as micro air vehicles (MAVs) (Shyy et al. 1999). For the MAV applications, membrane wings are preferred because of their inherently light-weight nature as well as variable camber feature. Potential flow (Newman and Low 1984; Newman 1987; Greenhalgh et al. 1984; Jackson and Christie 1986), laminar flow (Smith and Shyy 1995), and turbulent flow (Smith and Shyy 1996) calculations were used to simulate the equilibrium shapes in steady conditions. It was found that the potential flow theory can be sufficient only for small incidences and for membrane airfoils with small excess length ratios. For larger excess lengths and incidences, viscous effects and flow separation need to be included for accurate simulations in steady flow. Unsteady aspects of the fluid–structure interaction received less attention. In many applications, unsteady aspects of membrane wings are as important as their static performance. Both in nature

and small aircraft, flow separation and reattachment are typical and potential sources of unsteadiness. Yet, the unsteady aspects of the fluid–structure interactions of membrane airfoils at low Reynolds numbers are largely unknown.

The emphasis in this work is on the low Reynolds number flow over membrane airfoils. Membrane wings are often used in MAV applications, where the vehicles operate at Reynolds numbers below 10^5. Fixed-wing MAVs fly at relatively high angles of attack and close to the stall conditions due to the poor lift. Rise in unsteadiness and loss of control power are potential problems in low Reynolds number aerodynamics. Due to potentially damaging control problems, understanding of the gust response and stall conditions for fixed-wing MAVs is important. Hence, unsteady aspects of aerodynamics of membrane wings are far more important and critical for the performance and control of MAVs. For example, MAVs must be able to operate in gusts and unsteady free stream conditions (Shyy et al. 1997). They may also be subject to the instabilities caused by the fluid–structure interactions of the membrane wings. Even for membrane wings operating in steady free stream, aeroelastic instabilities may limit their operating envelope. At low incidences, the flow is mostly attached even though there might be a leading-edge separation bubble. Even these mostly attached flows may cause aeroelastic instabilities. At higher angles of attack, the flow is partially or fully separated over the membrane wings. These separated flows are expected to interact with the membrane structure.

Unsteadiness due to separation might cause buffeting of the membrane structure, and even may couple with the aeroelastic instabilities (Gordnier 2008; Visbal et al. 2008). In a recent study (Galvao et al. 2006), it was reported that standing waves with large mode numbers are observed for a membrane wing with aspect ratio of AR = 0.92. In a subsequent study (Song and Breuer 2007), similar vibrational modes were reported for $Re \geq 80{,}000$ for a rectangular wing with AR = 1.38. The flow is highly three-dimensional and tip vortices are dominant for low aspect ratio wings. It is possible that membrane vibrations can excite the separated shear layer forming the tip vortices and promote reattachment (Gursul et al. 2005). For a flexible nonslender delta wing, this mechanism delays the stall and increases the lift (Taylor et al. 2007). Hence, fluid–structure interactions on low aspect ratio wings are expected to be very complex. For this reason, we focus on two-dimensional membranes in this paper. The main objectives are to understand the unsteady aspects of these flows, membrane oscillations, and the interaction of the separated flows with the membrane. These aspects are investigated by using a high-speed camera for measurements of membrane shape and by time-accurate velocity measurements.

2 Experimental setup and methods

The experiments were carried out in the low-speed, closed-loop open-jet wind tunnel with the circular working section of the 760 mm in diameter, located in the Department of Mechanical Engineering, University of Bath. The membrane wing was placed in the test section by means of a frame and end plates as shown in Fig. 1a. Both the leading-edge and trailing-edge of the wing were fixed to circular plates at both ends to adjust the angle of attack as shown in Fig. 1b. The rigid leading- and trailing-edges run along the whole span of the test section between the two end plates. There is a small gap (1 mm) between the membrane and the end plates. The main feature of this setup is that the time-averaged membrane deformation is essentially two-dimensional, which can be seen in Fig. 1b. However fluctuating displacement might be three-dimensional due to the end effects. The wing had a span of 450 mm and chord length of $c = 150$ mm. The experiments were carried out over angles of attack from 0° to 30°, free stream velocities of 5, 7.5, and 10 m/s which correspond to Reynolds numbers, based on the chord length, of $Re = 53{,}100$, 79,700 and 106,000 respectively.

The membrane wing was made from a black latex rubber sheet with a thickness of $t = 0.2$ mm, Young's

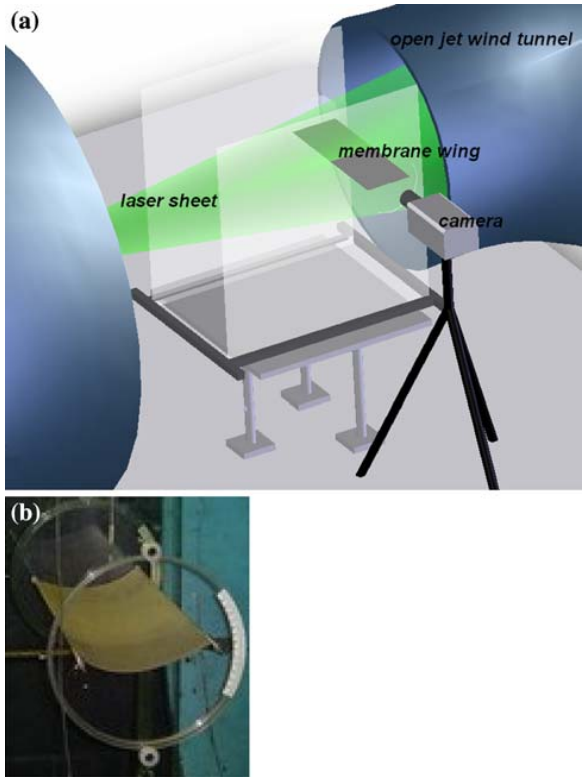


Fig. 1 **a** Schematic of experimental setup, **b** membrane airfoil in the test section

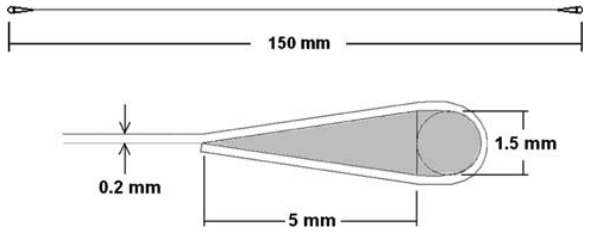


Fig. 2 Cross-section of the membrane airfoil and details of leading- and trailing-edge

modulus of $E = 2.2$ MPa, and density of $\rho_m = 1$ g/cm^3. The membrane was attached to the airfoil-shaped leading- and trailing-edges as shown in Fig. 2. The edges were designed with consideration that they must be strong enough to resist bending but small enough to produce a thin airfoil cross-section. The aeroelastic parameter used by Smith and Shyy (1996) is $\Pi = (Et/qc)^{1/3}$, where q is the free stream dynamic pressure. This parameter has a value of $\Pi = 5.77$, 4.41 and 3.64 corresponding to the free stream velocities $U_\infty = 5$, 7.5 and 10 m/s. The maximum camber took values up to around 10% of the chord length in present experiments. These values are consistent with the observations for bats (Song et al. 2008).

In addition, a rigid membrane airfoil (equivalent rigid cambered airfoil, i.e. with shape equivalent to the mean membrane deformation) was manufactured and tested in order to compare with the flexible membrane airfoil. In an effort to make a reasonable comparison over a range of incidences, the rigid wing was made of 1 mm thick stainless steel with a profile matching that of the average (over incidences) shape of the membrane wing at 5 m/s free stream velocity and at angles of attack from 10–18°. Also, the leading-edge and trailing-edge were rounded to give a similar shape to that of the flexible membrane.

In order to measure the membrane shape, a thin laser sheet from a 4-W Argon–Ion laser was used to illuminate the membrane surface in the streamwise plane located at the mid-span of the wing. A high speed camera (*Photron FASTCAM APX*) was positioned normal to the flow and images were captured at a rate of 1,500 frames per second. The resulting image appears as a white curve as shown by the example in Fig. 3. Present experimental technique can only be used to detect the chordwise modes. Spanwise modes may be present when the chordwise mode number increases, but this aspect has not been investigated with the present technique. The images with 1,024 × 1,024 pixel resolution in TIF format were obtained in 2 s, resulting in 3,000 images per each angle of attack and speed. Finally, the images were digitized through MatLab (Image Processing Toolbox) to find the coordinates. Two examples of the digitized membrane shapes at different instants are shown in Fig. 4. It is seen that the second mode and fifth

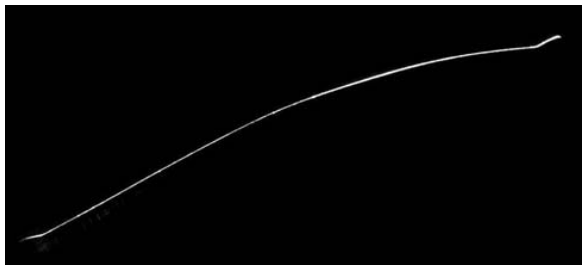


Fig. 3 Membrane shape as obtained from the laser sheet visualization, $U_\infty = 5$ m/s, $\alpha = 20°$. Flow is from *right* to *left*

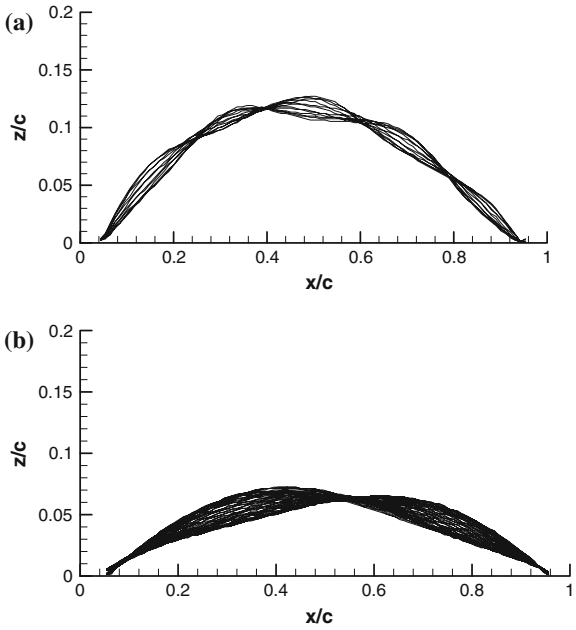


Fig. 4 Digitized membrane shape at different instants for **a** $U_\infty = 10$ m/s, $\alpha = 13°$; **b** $U_\infty = 5$ m/s, $\alpha = 28°$

mode of vibrations are captured in these examples. From the instantaneous coordinates of the membrane, the time-averaged membrane shape was calculated for each angle of attack and free stream velocity. In addition, the standard deviation of the membrane displacement was calculated as a function of the chordwise distance, which would easily indicate the mode of the vibrations. From the digitized membrane shapes as a function of time, it was also possible to study the time history of the displacement. An example of the time history of the displacement is shown in Fig. 5. The dominant frequency of the membrane oscillations can be found from the time history of membrane displacement. The dominant frequencies were found by spectral analysis using the Welch power spectral density method in MatLab.

Quantitative flow measurements were undertaken using a high frame-rate Digital Particle Image Velocimetry

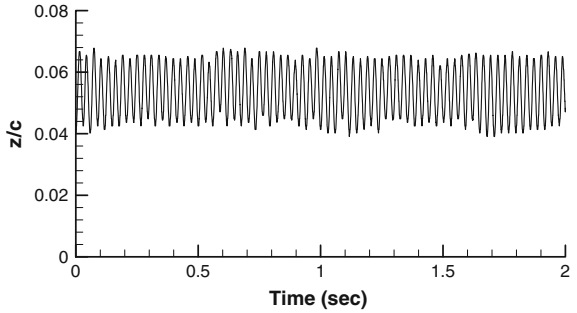


Fig. 5 Example of time history of membrane displacement as a function of time for $U_\infty = 5$ m/s, $\alpha = 28°$, $x/c = 0.32$

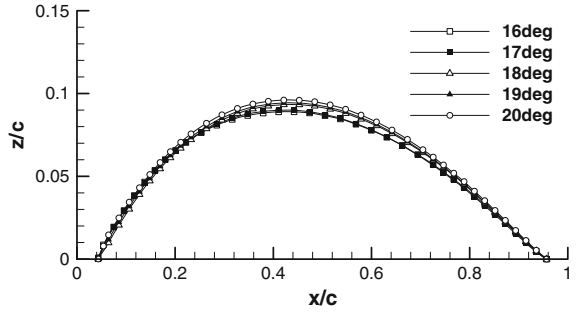


Fig. 6 Examples of time-averaged membrane shapes at different angles of attack, $U_\infty = 7.5$ m/s

(DPIV) system. Illumination of the desired plane was achieved using a New Wave Pegasus Nd:YLF double pulse high speed laser with a maximum energy of 10 mJ/pulse. The laser light sheet was placed parallel to the freestream velocity to illuminate a plane at the mid-span of the wing as shown in Fig. 1a. The PIV camera was placed normal to the flow, and the flow over the suction surface of the whole wing was imaged. In some experiments, specific areas near the leading-edge or trailing-edge were investigated. The images were captured using a TSI PowerView HS-3000 high speed CMOS camera. A TSI LaserPulse synchroniser unit was utilized to link the camera and the laser to enable the accurate capture for two frame cross-correlation analysis. For these measurements, the velocity field was captured at a rate of 1,500 frames per second, at a resolution of 1,024 by 1,024 pixels, producing sequences of 3,000 instantaneous velocity fields over 2 s. A TSI model 9307-6 multi-jet atomiser was used to produce oil droplets in order to seed the flow. The atomiser worked best using olive oil and the mean size of the droplets was 1 μm. The commercial software TSI Insight3G and a FFT cross-correlation algorithm were used for the analysis of the results obtained. The size of the interrogation window was 32 by 32 pixels. The effective grid size was around 2% of the chord length in these measurements. The measurement uncertainty for the velocity is estimated as 2% of the free stream velocity.

We also conducted flow visualization using a smoke-wire placed just upstream of the wing. In order to generate smoke, a strip of wire was clamped vertically, perpendicular to the flow at the mid-span of the wing. Oil was dropped to the wire continuously by a small valve on a dripping system. With a current applied to the wire, the oil was burnt by the heat, and the smoke was generated. The high speed camera (*Photron FASTCAM APX*) was used to capture images at a rate of 1,000 frames per second.

3 Results

3.1 Mean shape of membrane

Figure 6 shows examples of the time-averaged membrane shape for various angles of attack for $U_\infty = 7.5$ m/s. Note that the displacement is 0 in the small regions near the rigid leading-edge and trailing-edge. It is also seen in this plot that the maximum membrane displacement (also the point of maximum camber) is forward of the mid-chord point for high angles of attack, which is consistent with previous predictions and experiments (Greenhalgh et al. 1984). Figure 7a and b show the maximum camber and its

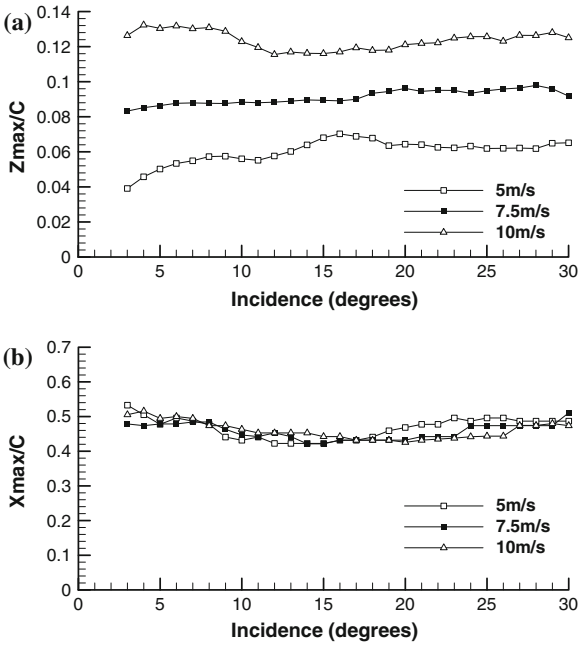


Fig. 7 Variation of **a** maximum camber, **b** chordwise location of the maximum camber as a function of angle of attack for three freestream velocities

chordwise location as a function of angle of attack for three different free stream velocities. Note that the data for $\alpha \leq 2°$ are not shown as the bistable instability known as luffing (Newman 1987) occurs in this region when the membrane is just as likely to set itself on one side as the other. The lift also experiences hysteresis (Greenhalgh et al. 1984) in this region. The maximum camber increases gradually up to for $\alpha = 16°$ for the lowest speed ($U_\infty = 5$ m/s) and then stays roughly constant. For $U_\infty = 7.5$ m/s, the maximum camber exhibits very small increase with angle of attack. For the largest free stream velocity, the variation in the maximum camber with incidence is relatively small. In general, the changes in the maximum camber are small as the angle of attack is varied for all three speeds, which is consistent with previous observations (Newman and Low 1984) and simulations (Gordnier 2008). This is different from what is expected in attached flows, and is due to the separated flows being predominant at these low Reynolds numbers.

Figure 7b shows that the location of the maximum camber is not very sensitive to the free stream velocity either. For all cases, the membrane shape is symmetrical about the mid-chord point for small incidences, which is also predicted theoretically (Newman and Low 1984). The location of maximum camber moves forward as the angle of attack is increased up to $\alpha \approx 15°$. This is also consistent with previous observations and predictions (Newman and Low 1984). There is a trend of the location of the maximum camber moving back to the mid-chord point at higher angles of attack.

3.2 Mean flow

Preliminary smoke flow visualization showed that the flow remains attached to the wing surface up to around 8° of angle attack for $U_\infty = 5$ m/s. In fact, a small leading-edge separation bubble was evident for both $U_\infty = 5$ m/s and $U_\infty = 7.5$ m/s. When the flow remains attached, oscillations of the membrane are very small. Consequently, we focus on larger angles of attack near the stall and in the post-stall regime.

Figure 8 shows the magnitude of the time-averaged velocity field superposed on the streamline pattern. For selected angles of attack, the data were taken in two

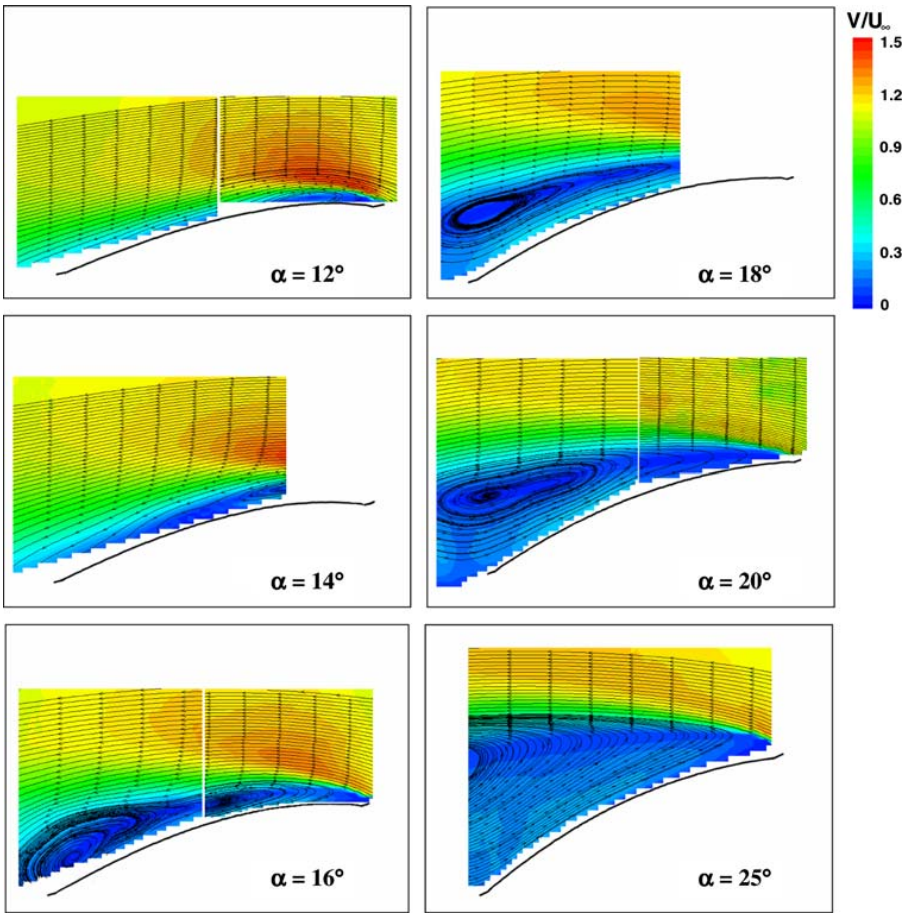


Fig. 8 Magnitude of the time-averaged velocity and streamlines at different angles of attack, $U_\infty = 5$ m/s. Flow is from *right* to *left*

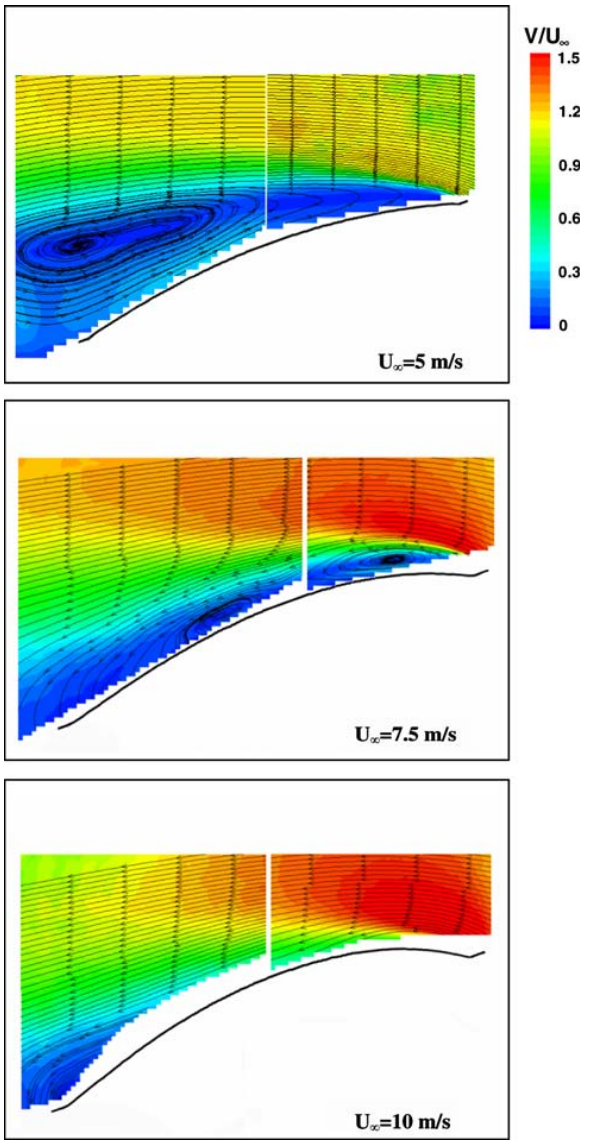


Fig. 9 Magnitude of the time-averaged velocity and streamlines at different freestream velocities, $\alpha = 20°$. Flow is from *right* to *left*

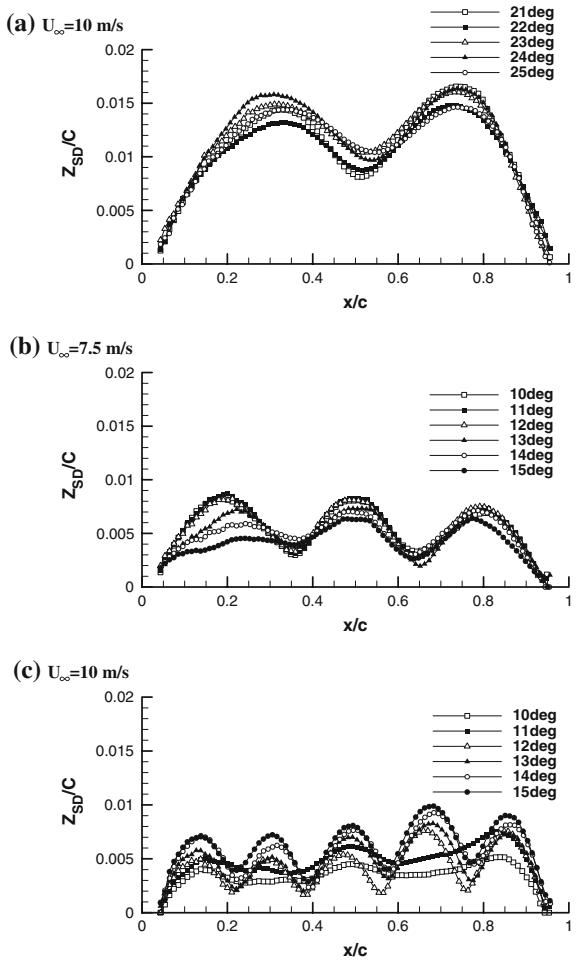


Fig. 10 Variation of the standard deviation of the membrane displacement normalized by chord length, showing examples for **a** second mode, **b** third mode and **c** fifth mode

different regions (one near the leading-edge and the other near the trailing-edge). Starting with $\alpha = 12°$, the flow is separated, but the shear layer remains very close to the membrane. With increasing angle of attack, the separated shear layer moves away from the surface. For $\alpha = 16°$, closed streamline pattern develops near the trailing-edge. The recirculation region becomes larger and moves further downstream with increasing angle of attack to $\alpha = 18°$ and 20°. For the largest angle of attack $\alpha = 25°$, the center of the closed recirculation region is no longer on the membrane wing. If the time-averaged membrane shape (Fig. 7) and the time-averaged flow (Fig. 8) are considered together, it is concluded that the changes in the mean membrane shape are much smaller than the changes in the mean flow as the angle of attack is varied for a given free stream velocity.

Figure 9 shows the magnitude of the time-averaged velocity and streamlines for different free stream velocities at $\alpha = 20°$. With increasing free stream velocity (increasing Reynolds number *Re* and decreasing aeroelastic parameter Π), there is a trend of shear layer getting closer to the wing surface while the camber of the wing increases. It will be shown later on that this affects the onset and the amplitude of the membrane oscillations. Although the effect of *Re* and Π could not be separated in these experiments, this was done in the numerical simulations by Gordnier (2008). It was shown that there was a significant reduction in the size of the separation zone with increasing Reynolds number at a fixed parameter Π.

3.3 Membrane oscillations

Figure 10 shows examples of different modes of membrane oscillations as indicated by the variation of the standard deviation with the chordwise distance. Examples of the second mode (Fig. 10a), the third mode (Fig. 10b), and the fifth mode (Fig. 10c) are shown for various free stream velocity and angle of attack combinations. Actually, it is noted for $\alpha = 10°$ and $11°$ in Fig. 10c that, the mode number appears to be three, and jumps to five for other incidences in this figure.

Figure 11 shows the variation of the mode number as a function of incidence and free stream velocity. An unambiguous identification of the second mode occurs first at the smallest angle of attack $\alpha = 11°$ for $U_\infty = 5$ m/s. With increasing free stream velocity, the smallest angle of attack at which a definite mode shape is identified decreases. This might be due to the unsteady shear layer getting closer to the wing surface (see Fig. 9). It is also noted that the mode number increases with the free stream velocity. It is interesting that, when $\alpha \geq 20°$, the second mode is observed regardless of the free stream velocity. This suggests a possible coupling of membrane with the wake flow at these post-stall incidences.

The dominant frequency as a function of angle of attack for three different free stream velocities is shown in Fig. 12. It is seen that, although the trend is similar to that of the mode number (Fig. 11), there are some differences. For example, for the same mode number of two (for $\alpha \geq 20°$), the measured frequencies are different for different free stream velocities. This is probably due to the change in the membrane shape, and therefore the tension. An attempt was made to predict the natural frequencies of the membrane, although it is not clear whether the linear elastic theory can be justified for this highly nonlinear problem. The membrane tension was estimated from the mean deformation and Young's modulus, and this was used to estimate the natural frequencies of the membrane based on the linear elastic theory. However, it should be kept in mind that this theoretical prediction is valid only for vacuum. The added mass effects as well as the effects of a mean flow are not taken into account. Substantial changes in the added mass are expected in separated flows (for an oscillating circular cylinder, the added mass may even become negative). Our prediction based on the linear elastic theory in vacuum, assuming pre-strain and two-dimensional membrane shape, provided frequencies roughly half of the experimental values. The membrane natural frequencies were predicted successfully for low aspect ratio membranes by using this approach (Song et al. 2008). However, Song et al. (2008) adjusted the value of the pre-strain (by adding additional strain of around 0.05) in order to fit to the theoretical model for the camber and argued that this value is within the fabrication uncertainty. Such a large pre-strain would correspond to nearly 7.5 mm difference in length in our setup, which we believe is not realistic. We believe that the difference between the linear theory and experiments highlight the nonlinear nature of the fluid–structure interaction.

The variation of the Strouhal number $\mathrm{St} = fc/U_\infty$ as a function of angle of attack is shown in Fig. 13. In general, the Strouhal number is in the order of unity. This is consistent with the findings of simulations for the same geometry but at much lower Reynolds number (Gordnier

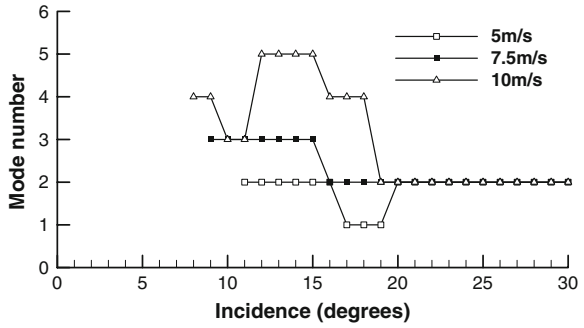


Fig. 11 Variation of the mode number as a function of incidence and freestream velocity

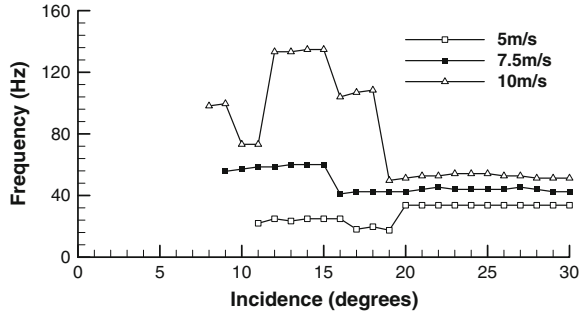


Fig. 12 Variation of the frequency of membrane oscillations as a function of incidence

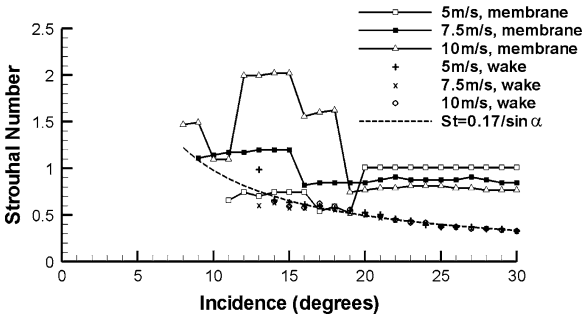


Fig. 13 Variation of Strouhal number of membrane oscillations as a function of incidence

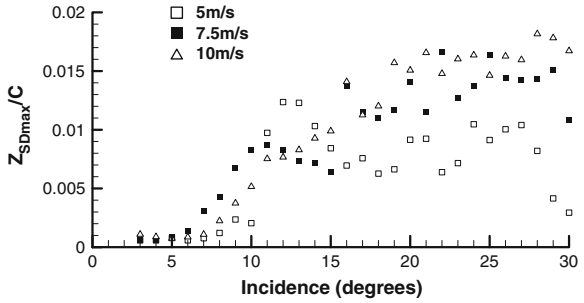


Fig. 14 Variation of the maximum standard deviation of membrane displacement normalized by chord length, as a function of incidence

2008). It is seen that, in particular, for $\alpha \geq 20°$, the Strouhal number is close to unity in Fig. 13. The Strouhal number increases with increasing free stream velocity at moderate incidences. Song et al. (2008) also reported increasing Strouhal number with free stream velocity for a low aspect ratio wing. An alternative definition of the Strouhal number, based on the vertical distance between the leading-edge and trailing-edge, is often used in the literature (Fage and Johansen 1927; Abernathy 1962):

$$\mathrm{St}_{\alpha} = \frac{fc \sin \alpha}{U_{\infty}}$$

This modified definition of the Strouhal number is reported to be constant for flat plates and thin airfoils, with the values given in the range of 0.16–0.22 (Fage and Johansen 1927; Abernathy 1962; Miranda et al. 2005). Vortex shedding frequency in the wake of the rigid membrane (equivalent rigid cambered airfoil) was measured with hot-wire anemometry and added to Fig. 13. It is seen that measured Strouhal numbers are not very different from those of flat plates and airfoils. Also, the dashed line shows $\mathrm{St}_{\alpha} = 0.17$, which agrees very well with our measurements at the three free stream velocities. This figure suggests that there might be coupling of the membrane oscillations with the vortex shedding from the membrane. In particular, for $\alpha \geq 20°$, where the second mode is always observed, the membrane oscillations might be coupled with the first harmonic of the natural frequency. In active flow control studies with rigid airfoils (Miranda et al. 2005; Wu et al. 1998), it was found that vortex shedding may lock on the first harmonic or the subharmonic of the natural frequency.

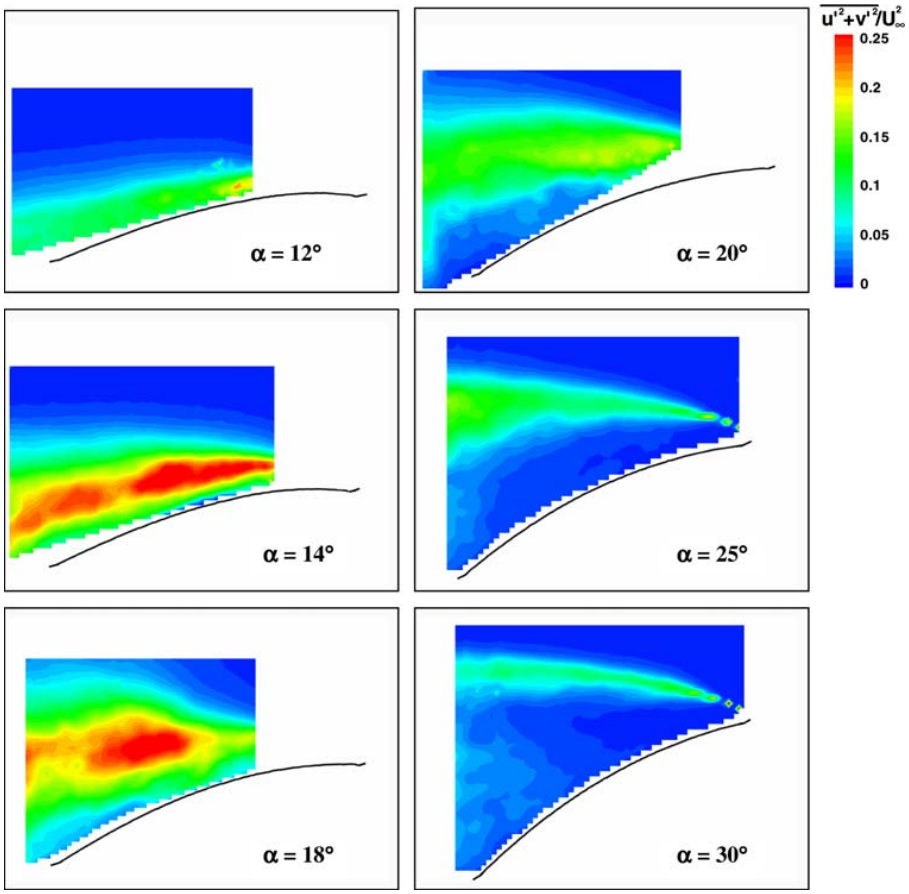


Fig. 15 Turbulence intensity at different angles of attack, $U_{\infty} = 5$ m/s. Flow is from *right* to *left*

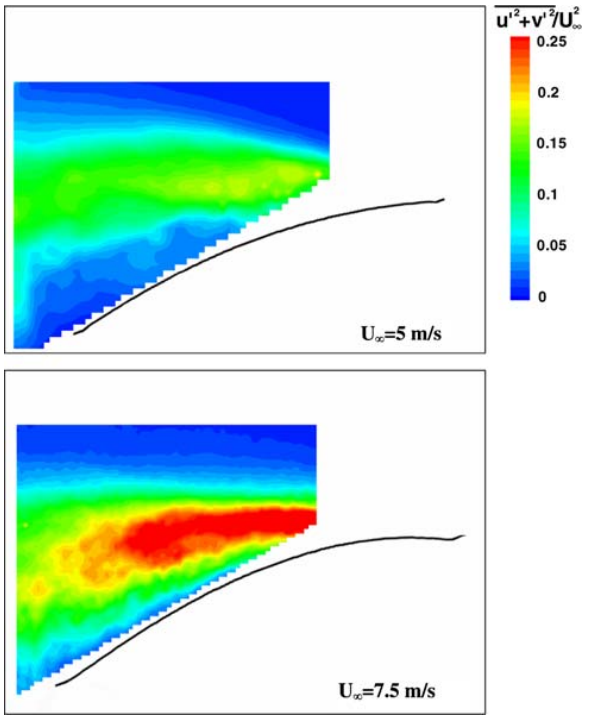


Fig. 16 Turbulence intensity at $\alpha = 20°$. Flow is from *right* to *left*

Finally, the maximum standard deviation of the membrane oscillations is shown as a function of angle of attack in Fig. 14. The amplitude of the oscillations starts to increase rapidly at a certain angle of attack and this incidence decreases with increasing free stream velocity. A local maximum is observed for $\alpha = 10$–$12°$, followed by increases or decreases depending on the free stream velocity and the mode jumps. For the lowest free stream velocity (and largest Π), the amplitude of the oscillations is somewhat smaller at high incidences. However, for the higher free stream velocities (and smaller Π), the amplitude has an increasing trend at high angles of attack. This might be partly due to the shear layer getting closer to the wing surface at higher free stream velocities.

3.4 Unsteady flow

Figure 15 shows the turbulence intensity $\overline{u'^2 + v'^2}/U_\infty^2$ in the measurement plane at different angles of attack for $U_\infty = 5$ m/s. The shear layer fluctuations move away from the surface with increasing angle of attack and also become stronger. Note that, for this free stream velocity, the largest membrane oscillations occur at $\alpha = 12°$ (see Fig. 14) for which the shear layer is closest to the surface as seen in Fig. 15. The velocity fluctuations become weaker and move away from the surface at very large incidences (see, for example, $\alpha = 30°$) and the membrane oscillations decrease.

Figure 16 shows comparison of the turbulence intensity when the free stream velocity is increased from $U_\infty = 5$–7.5 m/s for $\alpha = 20°$. It is seen that the shear layer becomes closer to the surface (which is also seen in the time-averaged flow in Fig. 9) while the magnitude of the fluctuations also increase. This explains the earlier onset of the membrane oscillations and also larger membrane oscillations at higher free stream velocities (see Fig. 14).

Figure 17 shows the Reynolds stress component as a function of angle of attack for $U_\infty = 5$ m/s. The peak in the Reynolds stress occurs along the shear layer where the time-averaged vorticity is maximum. The location of the maximum values of the Reynolds stress moves away from the membrane surface and downstream with increasing angle of attack. Peak magnitudes are observed for $\alpha = 14$–$18°$ range, then the peak value decreases at higher angles of attack. The relatively large value of the Reynolds stress observed for moderately high incidences is related to the formation of coherent structures. Examinations of the instantaneous flow fields suggest that the peak of the Reynolds stress coincides with the roll-up of the large vortices as shown in Fig. 18 for $\alpha = 18°$.

3.5 Flow–membrane coupling

Instantaneous velocity field measured by the high speed PIV system was examined to study the coupling of the instantaneous flow with the membrane deformation. In order to demonstrate the high degree of correlation between the unsteady flow and membrane shape, both the location of the shear layer (identified as the location of maximum vorticity) and the membrane displacement were analyzed for each instantaneous flow over several cycles of membrane oscillations. The inset in Fig. 19 shows the definition of the distance between the membrane surface and the shear layer, l, and the vertical displacement of the membrane from the time-averaged shape, z', measured at a chordwise location of 0.75c. Figure 19a shows the variation of the location of the shear layer and membrane displacement as a function of time for $\alpha = 13°$ and $U_\infty = 5$ m/s. It is seen that the location of the shear layer is highly correlated to the membrane oscillations. When the membrane displacement is maximum, the shear layer is farthest away from the wing at this chordwise location. Figure 19b shows instantaneous flow fields at instants corresponding to the minimum and maximum displacements of membrane at 0.75c. For this angle of attack, the shear layer flapping is significant, which might be the cause of the largest membrane oscillations for this free stream velocity.

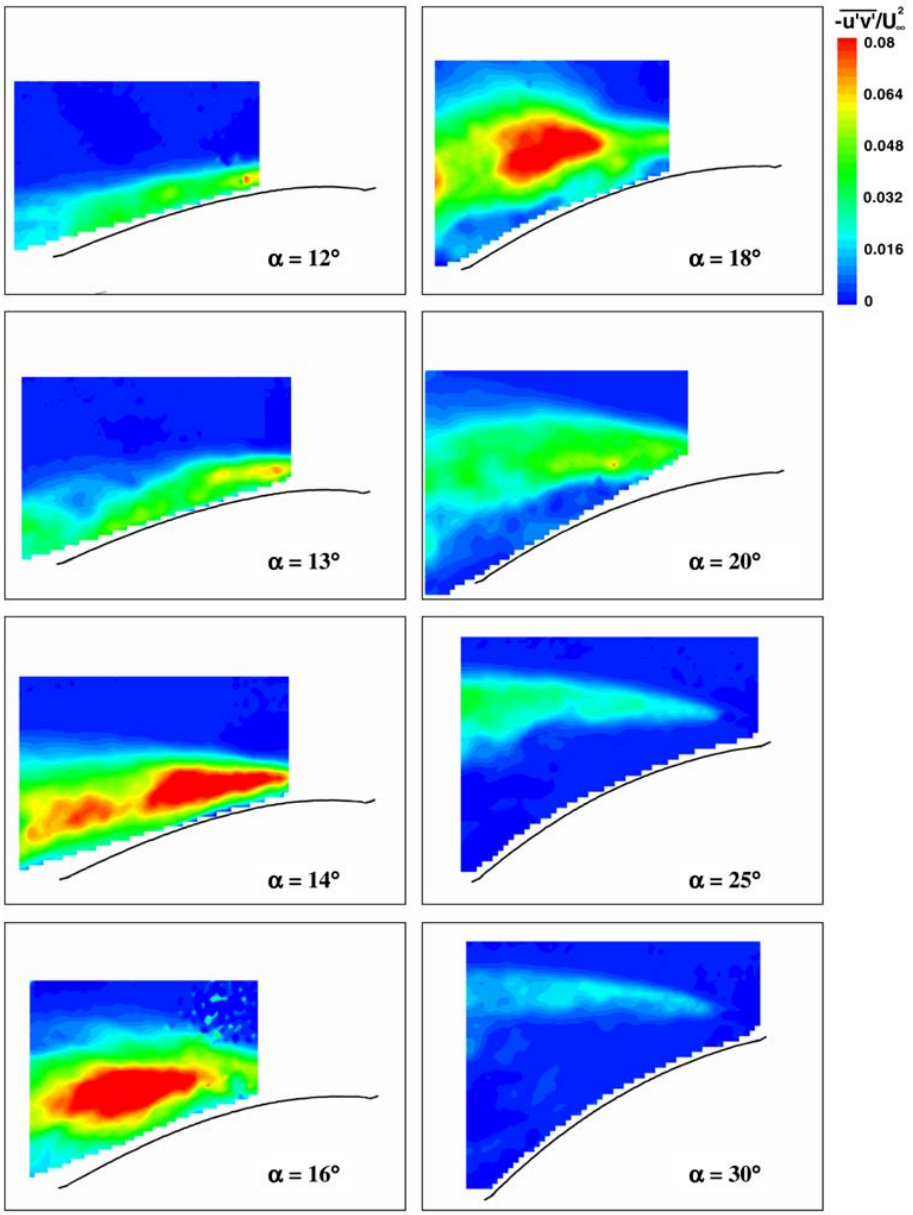


Fig. 17 Reynolds stress at different angles of attack, $U_\infty = 5$ m/s. Flow is from *right* to *left*

Figure 20 shows similar data for a higher angle of attack ($\alpha = 18°$) and $U_\infty = 5$ m/s. Positive correlation between the shear layer location and membrane displacement at 0.75c is similar to the case of $\alpha = 13°$. Figure 20b shows two instantaneous flow fields at instants of minimum and maximum membrane displacements. At this angle of attack, the roll-up of large vortices is more evident (see also Fig. 18).

3.6 Comparison of rigid and flexible membranes

Figure 21 shows smoke flow visualization with a high speed camera for the flexible membrane (left column) and the rigid membrane (right column). For the rigid membrane (a curved metal sheet), the mean camber is the same as for the flexible membrane, but of course there is no vibration. This presence or absence of membrane vibrations affects

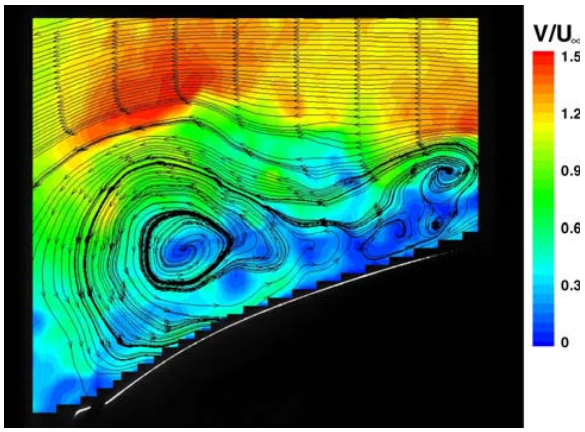
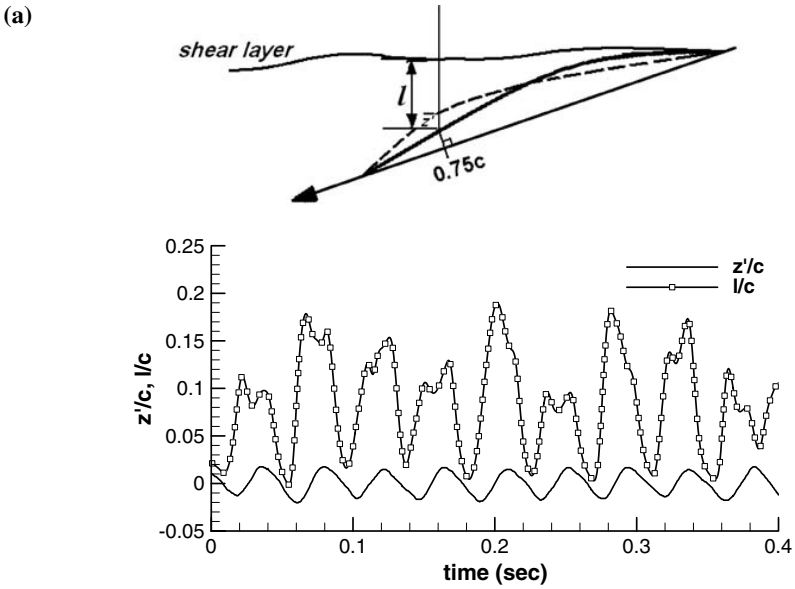


Fig. 18 Instantaneous velocity magnitude and streamlines for $U_\infty = 5$ m/s and $\alpha = 18°$. Flow is from *right* to *left*

the separated flow significantly. It is seen that for all incidences, the size of the separation and recirculation region is smaller for the flexible membrane. Also, at higher incidences ($\alpha = 18°$ and 20°), the roll-up of large vortices is evident over the flexible membrane whereas the shear layer does not exhibit large vortices for the rigid membrane. The oscillations of the membrane excite the shear layer, resulting in the roll-up of the large vortices. As the location of the shear layer (distance from the wing surface) is smaller for the flexible membrane, this suggests that drag might be smaller and stall might be delayed compared to the rigid membrane. Figure 22 shows the magnitude of the Reynolds stress for the flexible and rigid membrane wing for $\alpha = 18°$ and 20°. It is seen that the Reynolds stress is much larger for the flexible membrane due to the membrane vibrations, which excite the shear layer and promote the roll up of the vortices.



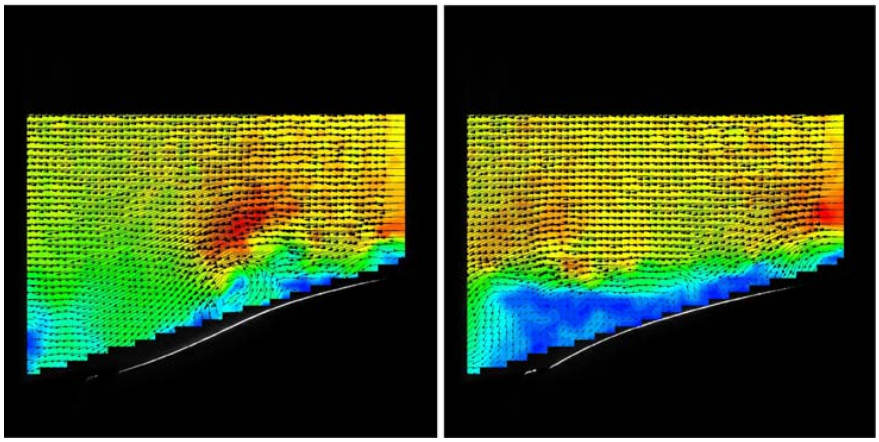


Fig. 19 **a** Time history of the locations of the shear layer and membrane as measured at 0.75c (the inset shows the definition of these quantities), **b** instantaneous flow field at minimum (*left*) and maximum (*right*) values of membrane displacement, $U_\infty = 5$ m/s, $\alpha = 13°$. Flow is from *right* to *left*

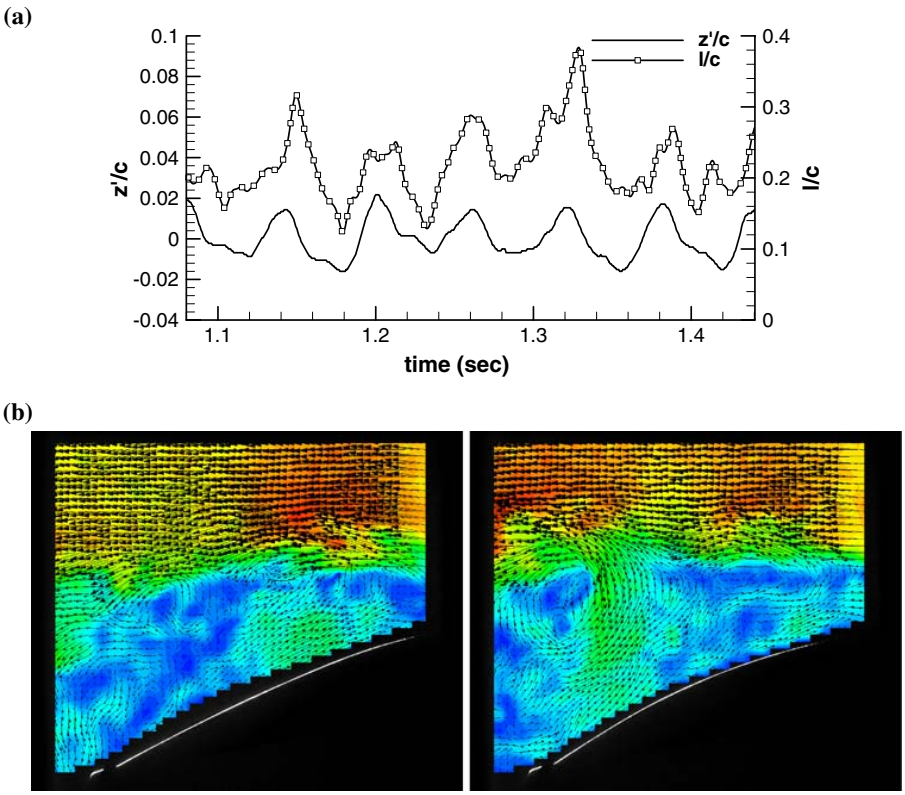


Fig. 20 **a** Time history of the locations of the shear layer and membrane as measured at 0.75c, **b** instantaneous flow field at minimum (*left*) and maximum (*right*) values of membrane displacement, $U_\infty = 5$ m/s, $\alpha = 18°$. Flow is from *right* to *left*

A similar comparison of flexible and rigid membranes was made in the simulations by Gordnier (2008) at $Re = 2{,}500$. It was shown that the dynamic motion of the membrane excites the shear layer causing it to roll up sooner and to form a series of vortices. This resulted in an increase in lift and decrease in drag.

4 Conclusions

Unsteady aerodynamics and fluid-structure interaction of a nominally two-dimensional membrane airfoil were investigated experimentally. Measurements of membrane shape, measurements of velocity field with a high speed PIV system, and flow visualization were performed. Experiments were conducted for three different values of the aeroelastic parameter $\Pi = (Et/qc)^{1/3}$ and in a Reynolds number range of $Re = 53{,}100$–$106{,}000$.

Mean membrane shape is not very sensitive to the changes in angle of attack, even though the time-averaged flow differs considerably: attached flow at small incidences, weakly separated flow at moderate incidences with the shear layer remaining close to the surface, and massively separated flows with large recirculation regions at high incidences. When the flow remains attached, the membrane oscillations are negligibly small. With increasing angle of attack, the amplitude of the oscillations increases rapidly around $\alpha = 10°$ for the lowest Reynolds number. The largest membrane oscillations occur when the separated shear layer remains close to the surface. With increasing angle of attack, the shear layer moves away from the surface. With increasing free stream velocity (increasing Reynolds number and decreasing aeroelastic parameter Π), the camber of the wing increases while the separated shear layer becomes more energetic and closer to the surface. This affects the onset of the vibrations and increases the amplitude.

Membrane oscillations with mode numbers of up to five were observed in the present experiments. The mode number increases with the free stream velocity at moderate incidences, whereas the second mode is always observed regardless of the free stream velocity for $\alpha \geq 20°$. The variation of the Strouhal number suggests a possible coupling of membrane oscillations with the Karman vortex shedding phenomenon in the wake. In particular, for $\alpha \geq 20°$, the data suggest possibility of the membrane oscillations locking on the first harmonic of the natural frequency.

Coupling of the membrane oscillations and separated shear layer has been demonstrated with the high degree of correlation between the membrane displacement and location of the shear layer from the analysis of instantaneous

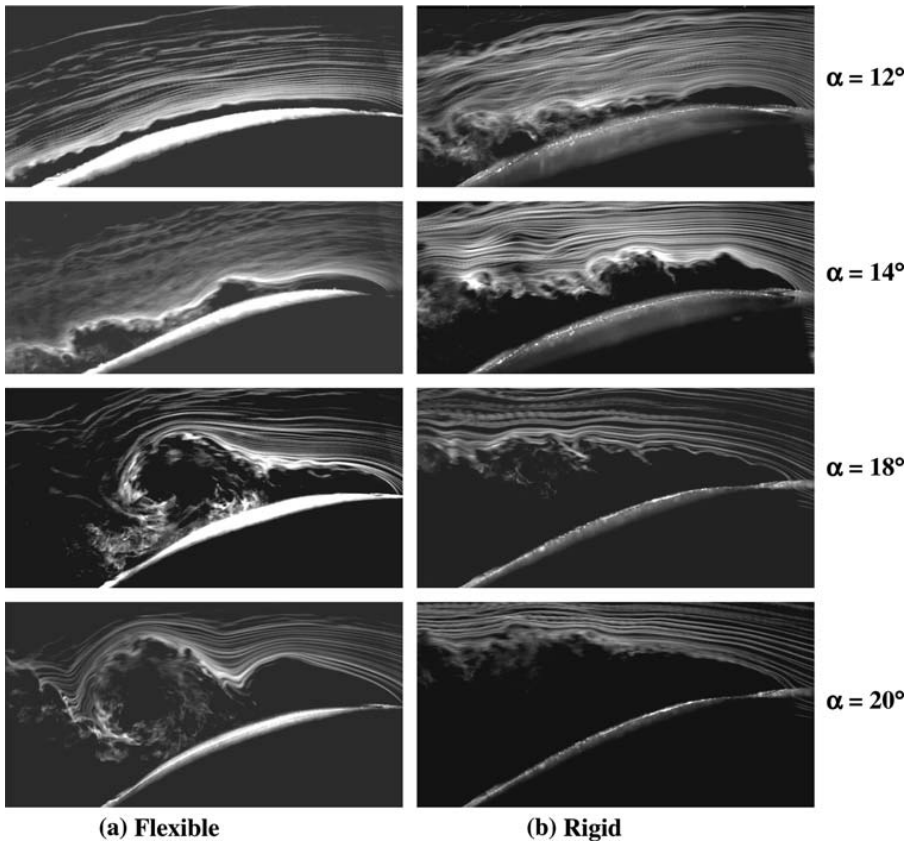


Fig. 21 Smoke flow visualization with a high speed camera at different incidences for **a** flexible, **b** rigid membrane, $U_\infty = 5$ m/s. Flow is from *right* to *left*

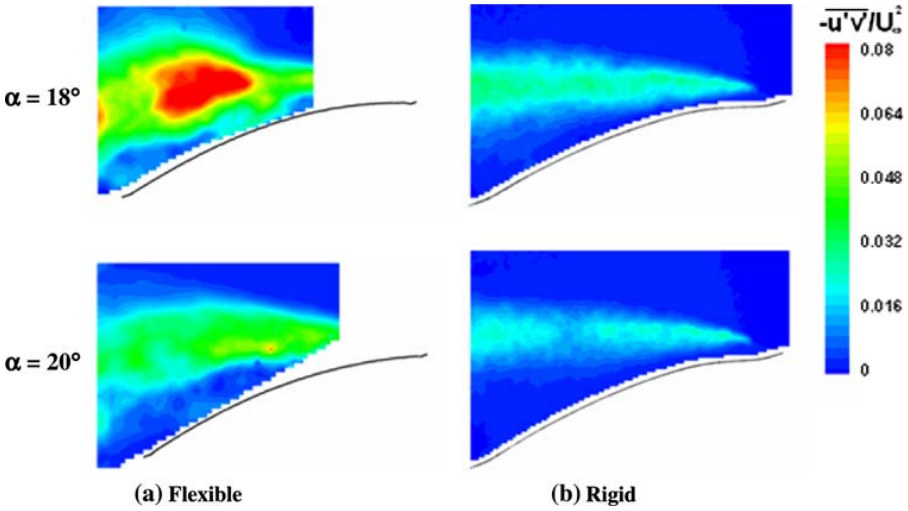


Fig. 22 Magnitude of the Reynolds stress for **a** flexible and **b** rigid membranes, $U_\infty = 5$ m/s. Flow is from *right* to *left*

flow fields. At moderate incidences, the flapping of the shear layer is more dominant, whereas at larger incidences ($\alpha = 16$–$20°$), the roll-up of large vortices is more characteristic of the flow. At even higher angles of attack ($\alpha = 25°$ and $30°$), the shear layer has relatively low level of fluctuations and does not exhibit the roll-up of the large vortices over the wing.

Comparison of the rigid and flexible membranes indicates that the size of the separation region is smaller for the flexible membrane. The oscillations of membrane excite

the shear layer, resulting in the roll-up of the large vortices over the wing. This suggests that membrane flexibility might decrease the drag and delay the stall.

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